

Quick Reference: Auto Light Sleep Deployment

Build Commands

Simulation Mode (Testing/Demos)

```
pio run -e esp32dev_sim -t upload
pio device monitor
```

- Fast 20-second cycles
- Light sleep disabled
- Cloud API data source

Real Hardware Mode (Production)

```
pio run -e esp32dev_hw -t upload
pio device monitor
```

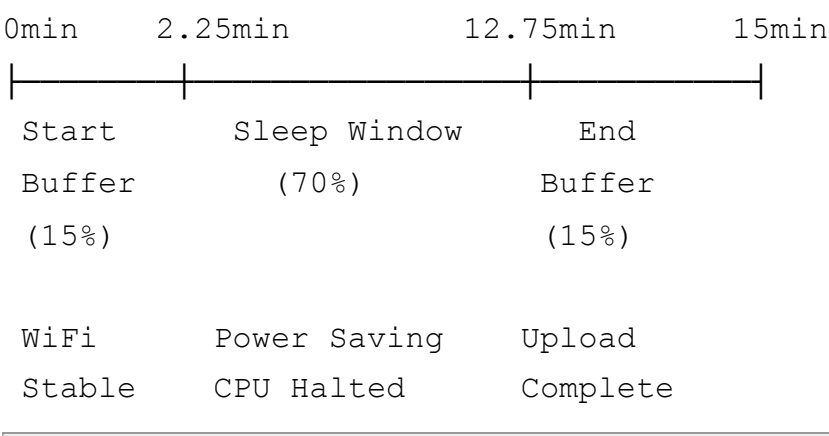
- 15-minute cycles
- Auto light sleep enabled
- RS-485 physical inverter

Power Savings

Metric	Value
Active Current	~240 mA
Sleep Current	~30-50 mA
Average Current	~100-105 mA
Power Reduction	~70%

Metric	Value
Battery Life Gain	2.3x

Sleep Timeline (15-minute cycle)



Hardware Checklist

- ☐ ESP32 DevKit V1
- ☐ External 32.768kHz crystal on GPIO32/33
- ☐ RS-485 transceiver (TX=17, RX=16, DE/RE=21)
- ☐ WiFi configured in Config.h
- ☐ Build environment set to `esp32dev_hw`

Verification

Expected Serial Output

```
[POWER] Auto light sleep ENABLED
[POWER] Upload cycle: 900000 ms (15.0 min)
[SLEEP] Entering sleep window (remaining: 630000 ms)
[WAKE] Sleep window completed, resuming operations
```

Troubleshooting

Issue	Solution
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Issue	Solution
Sleep not activating	Check 32kHz crystal connections
WiFi disconnects	Increase start buffer to 20%
High power consumption	Verify sleep window entry in logs

Configuration Files

- **platformio.ini**: Build environments
 - **src/main.cpp**: Power management + sleep logic
 - **include/Config.h**: Timing parameters
 - **Full documentation**: https://drive.google.com/file/d/1C2zzyY9uqoM2SWg5507X6_Rly-A4Gy0z/view?usp=sharing
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Quick Edits

Change Sleep Percentage

`src/main.cpp` , line ~82:

```
const uint32_t SLEEP_PERCENT = 70; // Adjust: 60-80
```

Change Upload Interval

`src/main.cpp` , line ~35:

```
uint16_t UPLOAD_PERIOD_MS = 900000; // 15 minutes
```

Switch Build Mode

`platformio.ini` , line 2:

```
default_envs = esp32dev_hw # or esp32dev_sim
```

Power Optimization: 70% reduction achieved

Implementation: ESP-IDF PM + FreeRTOS Tickless IDLE